

Earth's Future

RESEARCH ARTICLE

10.1029/2025EF007585

Key Points:

- Use of regional averages can miss extremes in outcomes for lower-income groups
- Income levels predominantly determine the relationship between joint resource security outcomes
- Regional drivers resulting in extreme joint resource security scenarios are discovered

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

G. J. Kim,
gkim3@tulane.edu

Citation:

Kim, G. J., O'Neill, B., Morris, J., Wise, M., Weyant, J., & Lamontagne, J. (2026). Identifying key uncertainties and drivers of future resource security outcomes through a multisector scenario ensemble. *Earth's Future*, 14, e2025EF007585. <https://doi.org/10.1029/2025EF007585>

Received 21 OCT 2025

Accepted 26 JUN 2026

© 2026 The Author(s). Earth's Future published by Wiley Periodicals LLC on behalf of American Geophysical Union. This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs License](#), which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Identifying Key Uncertainties and Drivers of Future Resource Security Outcomes Through a Multisector Scenario Ensemble

Gi Joo Kim¹ , Brian O'Neill² , Jennifer Morris³ , Marshall Wise² , John Weyant⁴ , and Jonathan Lamontagne⁵ 

¹ByWater Institute, Tulane University, New Orleans, LA, USA, ²Joint Global Change Research Institute, Pacific Northwest National Laboratory, College Park, MD, USA, ³Center for Sustainability Science and Strategy, Massachusetts Institute of Technology, Cambridge, MA, USA, ⁴Stanford University, Stanford, CA, USA, ⁵Department of Civil and Environmental Engineering, Tufts University, Medford, MA, USA

Abstract Food, energy, and water security are key criteria in analyses of complex human-Earth system interactions but are rarely analyzed jointly in scenario analysis of global change. Here, we generate a scenario ensemble centered around these outcomes using a global-scale multisector model that simulates critical interactions across multiple sectors. We consider a wide range of future uncertainties in both physical and societal factors, including population, economics, alternative scenarios of future carbon intensity, and land conservation. We evaluate each scenario by computing both regional and subregional resource security outcomes and identify the uncertainties and key drivers that lead to the outcomes of interest. We find that lower-income subgroups face higher burdens and more uncertain outcomes, and that drivers of resource security outcomes are heterogeneous across regions. Ultimately, our approach can help identify scenarios that can facilitate high-priority research on resource security outcomes, especially in multisector contexts.

Plain Language Summary Future changes in population, food production, and energy prices can create challenges that affect people's resource security. These challenges can impact the share of income that must be spent on food, household energy, or water, especially for those with lower incomes. Therefore, we need to explore how future resource security for different income groups may change under different conditions. We generate projections of how global changes, such as in population, gross domestic product, and technological advancement, may influence future living conditions. The results from this study show that both the living conditions and their uncertainty are different for people with lower incomes compared to those with higher incomes. Depending on regions and future assumptions, the share of income spent on food of the lowest income population can be as high as 49.6% compared to 5.5% of the highest income population's share. While income plays a major role in determining the relationship between regional food and energy affordability, other factors like geographic location and how energy or land is used also influence the joint outcomes. The findings from this study can help regional decision makers better understand how people's quality of life might change in the future and what factors influence it.

1. Introduction

Both global development frameworks, including the Sustainable Development Goals (Sachs, 2012), and individual studies increasingly underscore the need to address future resource security. Resource security refers to the reliability of access to and control over resources such as food, energy, water, and economic means. It is a key component of the broader concepts of human security (UNDP, 1994) and human well-being (Stiglitz et al., 2009). Multiple studies provide future projections of food security (Hasegawa et al., 2015; Obersteiner et al., 2016; Tilman & Clark, 2014; Van Meijl, Tabeau, et al., 2020), energy security (Bauen, 2006; Cameron et al., 2016; Pachauri et al., 2013), and water use sustainability (Wada & Bierkens, 2014), but they typically assess each outcome separately rather than jointly (Kling et al., 2017). Those that do tend to focus on aggregate outcomes such as food and energy production rather than indicators of resource security of the population (Van Vuuren et al., 2019). Furthermore, many of these studies adopt a single-sector or single-driver perspective and therefore do not fully capture the multi-sector interactions and feedback across human and Earth systems.

In addition, projections of resource security are often based on common scenario frameworks that are more broadly designed for different purposes, such as the Shared Socioeconomic Pathways-Representative Concentration Pathways (SSP-RCP scenarios) (O'Neill et al., 2020; Van Vuuren et al., 2014). Therefore, defining a set of scenarios explicitly designed to explore future security outcomes (i.e., outcome-driven community scenarios) is necessary, as argued by O'Neill et al. (2024). Defining such scenarios requires an understanding of the uncertainty in these outcomes, the synergies and tradeoffs that might exist between different dimensions of resource security, and the key drivers that determine outcomes of interest. We take a step in this direction by carrying out exploratory modeling to map the joint uncertainty in food, energy, and water security outcomes for a large number of global regions, taking into account uncertainty in multiple potential drivers.

Furthermore, previous literature shows that security outcomes demonstrate highly differentiated responses to external impacts across populations with different incomes (Sampedro et al., 2022a; Zhang et al., 2025). In other words, populations with high income are generally more resilient to external shocks, while lower-income subgroups are likely to experience more extreme impacts in their security outcomes. However, previous projections and scenarios have overlooked the importance of modeling multiple income subgroups, often focusing on regional or national averages that mask the differences across sub-populations with different income levels. Therefore, defining scenarios on future resource security outcomes, while differentiating across income subgroups, remains an important research gap that needs to be addressed.

To fill this gap, we generate a large ensemble of 3,888 scenarios and explore the outcomes across scales, including multiple sub-populations that are defined by income levels. These scenarios are generated by varying 12 uncertain factors related to socioeconomics, carbon intensity and land conservation, food, energy, water, and temperature impacts on water and energy. Our primary research objective in this study is to better understand the influence of future uncertainties on these outcomes across regions, sectors, and sub-populations. More specifically, we aim to answer the following research questions: (a) How will future human-Earth system uncertainties shape resource security outcomes in 2050 across regions, sectors, and income subgroups? (b) Which factors influence the relationships among joint security outcomes (e.g., synergy, tradeoff, or both possibilities)? and (c) What are the key drivers and pathways that lead to extreme resource security outcomes across regions, both separately and jointly?

To address these questions, we adopt exploratory modeling (Bankes, 1993) to develop a large ensemble of global change scenarios centered around the factors that directly influence resource security outcomes. Exploratory modeling is an analytical approach that uses multiple simulations from computational models to systematically investigate a wide range of plausible futures rather than trying to predict a single most likely outcome (Marchau et al., 2019). For the computational model, we leverage a leading global multisector model, which is a more detailed subclass of integrated assessment models that are widely used in global socioeconomic and energy system analyses (Weyant, 2014). Specifically, the model we adopt explicitly represents interactions across energy, water, land, socioeconomics, and Earth systems (Calvin et al., 2019), while representing multiple sub-populations differentiated by income levels (Sampedro et al., 2024). We then explore the resource security outcomes from the ensemble, highlighting the differences across sectors, regions, and income subgroups. Within this context, fully representing all dimensions of resource security remains challenging in a global multi-sector framework, motivating the use of tractable indicators to enable consistent cross-scenario comparison. Therefore, in this study, we demonstrate a methodological and exploratory study to analyze distributional and cross-sectoral interactions across a large scenario ensemble by using a set of tractable indicators to represent resource security outcomes. Finally, we adopt scenario discovery to identify the key factors driving scenarios of interest. Scenario discovery uses a classification algorithm to discover which combination of factors (e.g., socioeconomic growth, energy demand, and reservoir expansion) lead to extreme outcomes across the ensemble (Bryant & Lempert, 2010). Both exploratory modeling and scenario discovery approaches are especially salient in dealing with assumptions and scenarios that are explicitly non-probabilistic. Therefore, both approaches have been adopted in multiple research endeavors involving large ensembles of global change scenarios (Dolan et al., 2021).

The three key contributions from this study are as follows: (a) a thorough and systematic analysis to better understand future resource security outcomes across scales, (b) an identification of the key drivers and pathways leading to extreme security outcomes through scenario discovery, and (c) an extensive, searchable database of global change scenarios designed around resource security outcomes, which provides a rich resource for exploration across multiple research communities and demonstrates the scenario framework proposed by O'Neill

et al. (2024). This framework is designed around outcome-oriented scenario exploration, allowing researchers to examine which combinations of uncertain drivers lead to specific resource security outcomes, rather than focusing only on predefined socioeconomic or land conservation pathways.

2. Methods

To address our key research questions, we generate a large ensemble of 3,888 scenarios centered around resource security outcomes using a model that captures complex human and Earth systems interactions across scales: the Global Change Analysis Model (GCAM). GCAM is a global, dynamic-recursive partial equilibrium model that represents interactions of energy, water, land, socioeconomics, and Earth systems (Calvin et al., 2019). The ensemble is generated by varying 12 factors within 6 broader categories: (a) socioeconomics, (b) representative scenarios of future carbon intensity and land conservation, (c) food (agriculture), (d) energy, (e) water, and (f) temperature change impacts on energy and water (Figure S1 in Supporting Information S1). GCAM models how energy, water, land, socioeconomics, and Earth systems co-evolve in response to various sources of future uncertainties. The model operates across 32 geopolitical regions, and within each region the population is further divided into 10 income subgroups representing different portions of the income distribution. The model we adopt also includes a representation of 10 income subgroups in the food and residential energy sectors, which enables the modeling of security outcomes across income subgroups, including potentially vulnerable subpopulations within a region. As indicators of our security dimensions, we use food and residential energy burdens, defined as the fraction of income spent on food or residential energy services, and physical water scarcity, defined as the ratio of total withdrawals to renewable water supply. Food and energy burdens are useful but imperfect indicators of security; we discuss some limitations below (Section 2.3). While food and residential energy burdens are differentiated across income subgroups within a region, we compute physical water scarcity regionally. We explore the scenario ensemble outcomes using scenario discovery techniques, which enable us to identify the key drivers and pathways leading to scenarios with outcomes of interest.

2.1. Global Change Analysis Model (GCAM)

This study adopts a minted version of GCAM v7.1 (Ben Bond-Lamberty et al., 2024) to model future resource security outcomes of multiple sub-populations evolving under various future uncertainties. GCAM is an open source, global multisector model that captures the interactions between energy, water, land, socioeconomics, and Earth systems (Calvin et al., 2019). GCAM divides the world into 32 geopolitical regions, 235 water basins, and 384 land-use regions and adjusts the prices for hundreds of commodities for every 5-year time step from the base year 2015. GCAM uses a logit-choice algorithm (J. F. Clarke & Edmonds, 1993) combined with resource supply and cost curves to determine the equilibrium price at which the supply matches demand for each commodity. Due to GCAM's wide sectoral coverage and ability to model complex human-Earth system interactions, GCAM has been adopted for diverse research initiatives in multiple agencies. A comprehensive overview of the model, detailed explanations on the model structure, and details on each sector's representation in GCAM is included in the GCAM documentation (<https://jgcri.github.io/gcam-doc/index>). We include additional details on GCAM's food, energy, and water demand and supply in Text S1 in Supporting Information S1.

Specifically, we use the model with multiple income subgroup representation for the food and residential energy sectors, which enables the modeling of security outcomes of multiple sub-populations within a region (Sampedro et al., 2022b, 2024). In addition to general GCAM version's spatial decomposition (i.e., 32 geopolitical regions, 235 water basins, and 384 land-use regions), this version of the model further divides the population in each region into 10 subgroups differentiated by income levels (Sampedro et al., 2024). Each subgroup within each region has the same population (i.e., 10% of regional population for each subgroup), with different incomes determined by non-parametric income distribution projections (K. B. Narayan et al., 2023). Sub-populations in lower income subgroups (e.g., lower 10% income subgroup) are allocated with less income compared to the sub-population in higher income subgroups (e.g., upper 10% income subgroup). In this paper, we focus our analysis on the security outcomes for the sub-population with the lowest income, with a few examples highlighting the outcomes for higher income subgroups.

2.2. Scenario Ensemble Design

We use exploratory modeling (Bankes, 1993) to generate a large ensemble of searchable scenarios and apply scenario discovery (Bryant & Lempert, 2010) to explore the influence of future uncertainty on human well-being outcomes across multiple income subgroups. Because both exploratory modeling and scenario discovery involve scrutinizing the outputs from a large ensemble of scenarios, we design a large ensemble of scenarios for our analysis. To generate the ensemble, we consider a wide array of potential drivers that directly impact resource security outcomes. We sample the drivers based on a comprehensive literature review of previous experiments that utilize global change scenario ensembles (Dolan et al., 2021, 2022; Lamontagne et al., 2018; Niazi et al., 2024; Wessel et al., 2024; Woodard et al., 2023).

In this study, we sample 12 uncertain factors from six broader categories to widely capture the uncertain factors influencing future resource security outcomes. The six categories we include are: (a) socioeconomics, (b) carbon intensity and land conservation, (c) food (agriculture), (d) energy, (e) water, and (f) temperature impacts on water supply and energy demand. Partial factorial sampling of all factors results in a 3,888-scenario ensemble. Among the ensemble, GCAM was not able to find equilibrium solutions to 153 scenarios due to infeasible combinations of factors or because the solver exceeded the maximum number of iterations. Therefore, we perform our analysis on the remaining 3,735 scenarios (i.e., 96.1% of the ensemble), representing all feasible outcomes within the ensemble simulations. All uncertain factors varied in our ensemble are discussed in more detail below.

The first two factors belong to the socioeconomics category, which include population, the gross domestic product (GDP), and income distribution assumptions. We pair the population and GDP assumptions together and adopt GCAM's qualitative assumptions of Shared Socioeconomic Pathway scenarios (Calvin et al., 2017). The five SSPs correspond to plausible narratives on how the future will unfold based on the level of challenges to mitigation and challenges to adaptation. For instance, the SSP2 socioeconomics assumption assumes a “middle-of-the-road” future, with a global population of 9 billion people and GDP per capita of \$33,307 by 2100. In contrast, the SSP3 assumption represents a future with high population and low economic growth (i.e., 12.7 billion people with GDP per capita of \$12,092 in 2100), while the SSP5 assumption reflects a future with the opposite conditions (i.e., 7.4 billion people with GDP per capita of \$83,496 in 2100). The income distribution assumptions allocate regional GDP to 10 income subgroups for each simulation period. We adopt the GDP and income distribution assumptions for the SSP2, SSP3, and SSP5 scenarios. The income distributions are computed using a two-component principal component analysis by K. B. Narayan et al. (2024). While some socioeconomic assumptions may not perfectly align with the assumptions from other categories (e.g., high population growth assumption paired with land conservation scenarios), we consider them to explore a wide range of population and income trajectories and their implications for resource security outcomes.

The second category is the carbon intensity and land conservation category, which includes scenarios on global net carbon intensity and land conservation. For the global carbon intensity scenarios, we consider two cases as representative background scenarios to provide general insights: (a) the “increasing carbon” scenario in which fossil fuel use continues to grow and (b) the “decreasing carbon” scenario, which follows the assumptions from the “updated pledges-continued ambition” scenario from previous literature (Iyer et al., 2022; Ou et al., 2021). Under the decreasing carbon scenario, countries achieve long-term strategies, shorter-term pledges, and net-zero targets in target years, which are further maintained with a minimum rate of reductions afterwards. This results in a sustained trajectory rather than a single step decrease in the decreasing carbon scenario (Figure S2 in Supporting Information S1). For the assumption with no land conservation, we adopt the definition of protected land from the International Union for Conservation of Nature (Ravenel & Redford, 2009). According to Dolan et al. (2022), this assumption ultimately results in 26%–27% of global land being protected. In contrast, scenarios with the land conservation assumption define protected land as 90% of all unmanaged land, resulting in around 60% of global land being protected. We only couple the scenarios with the land conservation assumption with decreasing carbon scenarios.

The third category includes three factors that directly impact the food (or agriculture) sector. The first factor controls the level of agricultural trade in GCAM, by varying the preferences for domestically produced agricultural commodities and imported agricultural commodities. We adopt the low, medium, high trade scenarios developed by Graham et al. (2023) which we couple with the GDP and population assumptions (i.e., low trade with SSP3, medium trade with SSP2, and high trade with SSP5 GDP and population assumptions). The second factor we vary is the rate of agricultural productivity growth. For the three agricultural productivity growth

scenarios, we adopt the parameters corresponding to 50% of GCAM default, GCAM default, and 150% of GCAM default assumptions for low, medium, and high growth scenarios, respectively. The last factor in the food category is the food demand sensitivity assumption, which determines how consumer demand changes with respect to the changes in food prices. Here, we consider two possibilities: the reference sensitivity case and the higher sensitivity case (i.e., lower food demand under the same price compared to the reference sensitivity scenario). We represent consumer demand sensitivities by varying 11 parameters required for the food demand model used in GCAM (Table S1 in Supporting Information S1).

The fourth category includes two factors pertaining to the energy sector. The first factor reflects future buildings energy efficiency improvements with two discrete levels: (a) the “reference” case, in which the efficiency improvements are consistent with GCAM's default assumptions and (b) the “high efficiency” case with increased energy efficiency over time in the buildings sector, represented through higher heating and cooling efficiency improvements, reduced plug load, and reduced plug space (Gambhir et al., 2022). The second factor varies the demand sensitivity of consumers to residential energy price changes, similar to the food demand sensitivity assumption. Whereas the “reference” sensitivity scenario adopts the satiation level for residential building energy services consistent with GCAM default assumptions, the “high sensitivity” scenario reduces the satiation level from the reference sensitivity scenario by 10% (i.e., 90% of the reference sensitivity scenario's satiation level).

The fifth category relates directly to the rate of increase in future water resources infrastructure in GCAM, affecting water supply. Aligning with previous literature (Dolan et al., 2021; Turner et al., 2019), we consider two scenarios with different rates of future water storage infrastructure expansion in GCAM. The “restricted” scenario assumes that no additional water resources infrastructure is built after 2015. Under this assumption, the ratio of surface water to accessible water is kept constant up to 2100. In contrast, the “expanded” scenario assumes that all basins linearly increase access to surface water from the 2015 levels to maximum levels by 2100 (i.e., all accessible water is stored under the expanded scenario in 2100).

Finally, the sixth category includes temperature impacts on water and energy sectors, derived from three different earth system models (ESMs). Temperature impacts on food (agriculture) are excluded in this ensemble, as the assumptions on agricultural productivity growth rates cover a wider range that encompasses the potential impacts from temperature change. To reflect future temperature impacts on water supply and energy demand, we adopt runoff (Zhao et al., 2023) and heating and cooling degree days projections from the Inter-Sectoral Impact Model Intercomparison Project 3b (ISIMIP3b) (Lange & Büchner, 2021)'s downscaled and bias-corrected temperature data. They are then modeled through Xanthos (Li et al., 2017) and Helios (Zhao et al., 2024), exogenous models used to compute runoff and heating and cooling degree days in GCAM. In this study, we choose the forcing trajectories from three ESMs (i.e., GFDL-ESM4, IPSL-CM6A-LR, and CanESM5) and two radiative forcing levels (i.e., RCP 2.6 W/m² and RCP 7.0 W/m²). The RCP 2.6 scenario corresponds to the global radiative forcing levels reaching 2.6 W/m² by 2100 and is coupled with decreasing carbon scenarios. We couple the RCP 7.0 scenario with increasing carbon scenarios. We summarize all factors we include during our ensemble generation in Table 1. Note that we make no analysis about the likelihood of all uncertain factors in the ensemble, but we use them to generate representative scenarios to illustrate general findings under different future conditions.

2.3. Indicators for Resource Security Outcomes

This study explores resource security outcomes across income subgroups, regions, and scenarios using indicators for food, energy, and water security. We adopt the concepts of food burden (i.e., expenditure share on food), residential energy burden (i.e., expenditure share on residential energy services), and physical water scarcity (i.e., total water withdrawals relative to renewable water supply) to capture the broad aspects of resource security. Among our indicators, food and residential energy burdens measure each sub-population's security outcome in terms of economic affordability of essential goods and services. On the other hand, physical water scarcity captures another aspect of resource security related to hydrological sustainability. In our study, we assume that all indicators with lower values indicate higher food, energy, and water security outcomes.

The concept of food burden is a useful indicator of food security, and has been adopted as one of the food security indicators from the Food and Agriculture Organization of the United Nations (Lele et al., 2016). This indicator captures the portion of income a sub-population spends on food, which is an irreplaceable resource. Typically, food burden declines with rising income (the so-called Engel curve for food; Clements & Si, 2018). However, it is not an unambiguous measure of food security, especially because it does not directly measure the consumption

Table 1
Details on Each Uncertain Factor Included in Our Ensemble Design

Category	Uncertain factor	Number of discrete cases	Discrete cases modeled	Short description and explanation in GCAM
Socioeconomics	Population and GDP	3	SSP2, SSP3, SSP5	Paired population and GDP assumptions for SSP2, SSP3, and SSP5, consistent with the representations in GCAM (Calvin et al., 2017) Note that only population and GDP assumptions are varied here, rather than other qualitative assumptions across the SSPs
	Income Distributions	3	SSP2, SSP3, SSP5	Projected income distributions for SSP2, SSP3, and SSP5 scenarios using a two-component principal component analysis approach (K. B. Narayan et al., 2024)
Carbon Intensity and Land Conservation	Carbon Intensity	2	Increasing Carbon, Decreasing Carbon Scenarios	Increasing Carbon: a reference case with continued growth in fossil fuel use Decreasing Carbon: Global net carbon intensity decrease based on literature-based estimates (Iyer et al., 2022; Ou et al., 2021)
	Land Conservation	2	None, Land Conservation	None: defines protected land following the definition from the International Union for Conservation of Nature (Ravenel & Redford, 2009) Land Conservation: defines protected land as 90% of all unmanaged land (Dolan et al., 2022) Land conservation assumption is only coupled with scenarios with decreasing carbon scenarios
Food (Agriculture)	Level of Agricultural Trade	3	Low, Medium, High Trade	Low, medium, high trade scenarios developed for GCAM (Graham et al., 2023), which varies the preferences for domestically produced agricultural commodities and imported agricultural commodities Level of agricultural trade is coupled with GDP and population assumptions (i.e., low trade with SSP3, medium trade with SSP2, high trade with SSP5)
	Agricultural Productivity Growth Rate	3	Low, Medium, High Productivity	Low, medium, high agricultural productivity growth rate assumption, consistent with the representations in GCAM (Low: SSP3, medium: SSP2, high: SSP1)
	Food Demand Sensitivity	2	Reference, High Sensitivity	Set of 11 food demand parameters required for Ambrosia, the exogenous food demand model for GCAM.
Energy	Buildings Energy Efficiency	2	Reference, High Efficiency	Reference: Energy efficiency improvements consistent with GCAM core assumptions High Efficiency: Increased energy efficiency over time in the buildings sector (Gambhir et al., 2022) (i.e., higher heating and cooling efficiency improvements, reduced plug load, and reduced plug space)
	Energy Demand Sensitivity	2	Reference, High Sensitivity	Reference: Satiation values for residential building energy services consistent with GCAM core assumptions High Sensitivity: 90% of the satiation values from GCAM core assumptions
Water	Reservoir Expansion	2	Restricted, Expanded	Change in future expansion rates of water storage infrastructure in GCAM (Dolan et al., 2021; Turner et al., 2019) Restricted: Assumes fixed rate of base year surface water access until 2100 Expanded: All basins increase access to surface water linearly from base year levels to maximum levels by 2100

Table 1
Continued

Category	Uncertain factor	Number of discrete cases	Discrete cases modeled	Short description and explanation in GCAM
Temperature Impacts on Water and Energy	Earth System Models (ESMs)	3	GFDL-ESM4, IPSL-CM6A-LR, CANESM5	Projected runoff (Zhao et al., 2023) and heating and cooling degree days, reflecting future temperature impacts on water supply and energy demand.
	Radiative Forcing Levels	2	RCP 2.6, RCP 7.0	Downscaled and bias-corrected temperature data from the Inter-Sectoral Impact Model Intercomparison Project 3b (ISIMIP3b) (Lange & Büchner, 2021) are modeled through Xanthos and Helios, exogenous models used to model runoff and heating and cooling degree days for GCAM RCP 2.6 scenarios are coupled with scenarios with decreasing carbon scenarios; RCP 7.0 scenarios are coupled with scenarios with increasing carbon scenarios

itself. Food burden is affected by prices, not just income, so for example, a high burden may reflect higher prices rather than limited access. Food burden also does not reflect nutritional quality or consumption inequality and does not account for household food production (FAO, 2023). Nonetheless, it provides a useful starting point for illustrating the construction of scenarios based on multiple dimensions of resource security.

Food demand in GCAM is modeled separately for staple and nonstaple food commodities following the framework from Edmonds et al., 2017. Initially, demand for staple foods rises when incomes are low, reaches a maximum at roughly \$1,000 per capita, and then decreases as income continues to grow. On the other hand, demand for nonstaple food grows continuously with income, eventually saturating at higher income levels (K. Narayan et al., 2021) (Figure S4 in Supporting Information S1). Therefore, substitution between staple and nonstaple foods emerges endogenously as income rises and relative price changes, reflecting shifts in dietary preferences. Readers are referred to the Supporting Information S1 for details on the specific commodities included in the staple and nonstaple categories (Text S1 in Supporting Information S1) and the parameters used to model food demand in this study (Table S1 in Supporting Information S1).

In our study, we define food burden as per capita expenditure on food as a percentage of per capita income. Equation 1 defines the food burden of income subgroup, i , GCAM region, r , scenario, s , and year, y . To compute the food burden for each income subgroup, region, scenario, and year, we multiply each food commodity (f)'s consumption (FC_f) with its regional price (FP_f) and aggregate it across all food commodities in GCAM. We then divide the aggregated food expenditure by sub-population's income (I_i):

$$\text{Food burden}_{i,r,s,y} = \sum_f (FC_{f,i,r,s,y} \times FP_{f,r,s,y}) / I_{i,r,s,y} [\%] \quad (1)$$

where i, r, s, y , and f correspond to a given income subgroup, region, scenario, year, and food commodity, respectively.

In a more comprehensive assessment, food security would ideally be evaluated using a combination of complementary indicators that capture its multiple dimensions of availability, access, utilization, and stability (Van Meijl, Tabeau, et al., 2020). These include consumption-based measures (e.g., caloric adequacy relative to minimum dietary requirements), nutrition-sensitive indicators (e.g., dietary diversity), and indicators measuring temporal or adaptive responses (e.g., substitution between staple and non-staple food). Such indicators are widely used in multi-dimensional frameworks and provide a more holistic representation of food security outcomes than burden-based indicators alone.

Similarly to food burden, residential energy burden can capture the economic affordability of residential energy services for each income subgroup (Sampedro et al., 2024; Zhang et al., 2025). Energy burden typically follows an inverted-U shape with income, rising at low income levels and falling at higher income (Dorband et al., 2019). Similar to food burden, care needs to be taken in its interpretation. At low incomes, a low burden often reflects

lack of access, rather than high energy security, and as energy burden increases with increasing income, it is not purely a signal of decreasing security but includes increasing access and consumption that comes closer to sufficiency. Similarly to food security indicators, a more comprehensive assessment of energy security would combine residential energy burden with complementary indicators of availability, accessibility, and reliability to better capture its multidimensional nature. Despite its limitations, energy burden is a practical indicator for examining the affordability dimensions of energy security.

Based on GCAM's residential building energy consumption differentiation, we consider heating, cooling, and other services to compute the residential energy burden (Text S1 in Supporting Information S1). In our study, we compute residential energy burden by dividing the aggregated residential energy expenditure by sub-population income (I_i). The residential energy expenditure is computed by aggregating the product of residential energy service output (SO_e) and the service's cost (SC_e) across all residential energy services, e :

$$\text{Residential energy burden}_{i,r,s,y} = \sum_e (SO_{e,i,r,s,y} \times SC_{e,r,s,y}) / I_{i,r,s,y} [\%] \quad (2)$$

Finally, we adopt the definition of physical water scarcity from Vörösmarty et al. (2000) for our measure of water security. Physical water scarcity captures the degree of human interaction with respect to sustainable water supply as the ratio of total water withdrawals (TWW) to total renewable water supply (TWS). To compute total water withdrawals, we compute the sum of total water withdrawals from GCAM's six end-use sectors. Total renewable water supply includes water from surface water, renewable groundwater, reservoir storage, and baseflow excluding environmental flows for each GCAM basin, b (Birnbbaum et al., 2022). We compute physical water scarcity for each GCAM basin, b , scenario, s , and year, y :

$$\text{Physical water scarcity}_{b,s,y} = \text{TWW}_{b,s,y} / \text{TWS}_{b,s,y} [-] \quad (3)$$

This indicator represents exposure to physical water availability constraints, which may affect households through direct pathways such as physical constraints on irrigation water availability, groundwater depletion, and water quality impacts. Physical water scarcity therefore complements burden indicators, which reflect economic impacts propagated through price changes.

Furthermore, while GCAM represents water demand across multiple sectors, including agriculture, electricity generation, industry, and municipal use, municipal water demand is modeled at an aggregate regional level and is not disaggregated by household income groups. As a result, the model captures sectoral water demand dynamics but does not represent household-level municipal water consumption. Therefore, our analysis does not directly estimate household-level municipal water consumption or simulate how changes in water prices would affect water consumption for different income groups. Ideally, household-level indicators for water security would include measures of water affordability, access to safe water, and consumption adequacy (i.e., sufficient daily water intake). Therefore, physical water scarcity used in this study should thus be interpreted as reflecting basin-level water availability, rather than household-level municipal water affordability or access.

GCAM yields different spatial resolutions for the water sector outcomes (235 water basins) than the food and energy sector outcomes (32 geopolitical regions). Therefore, we adopt a crop revenue-based weighted-average aggregation method from Kim et al., 2025 to transform basin-level physical water scarcity into regional physical water scarcity, assigning higher weights to basins with higher agricultural revenue. For more detailed explanations of the crop revenue-weighted aggregation used in this study, readers are referred to Kim et al., 2025.

2.4. Classification and Regression Trees

Scenario discovery often adopts machine-learning-based classification algorithms to identify the combination of drivers leading to the outcomes of interest in large ensemble simulations. In this study, we adopt the Classification and Regression Trees (CART) algorithm (Breiman et al., 2017) to perform scenario discovery on resource security outcomes from our ensemble. Due to its wide applicability and high interpretability of outcomes, CART has been adopted in multiple studies using large ensemble simulations to identify the most important factors leading to the scenarios of interest (Birnbbaum et al., 2022; Dolan et al., 2021; Lamontagne et al., 2018; Wessel et al., 2024).

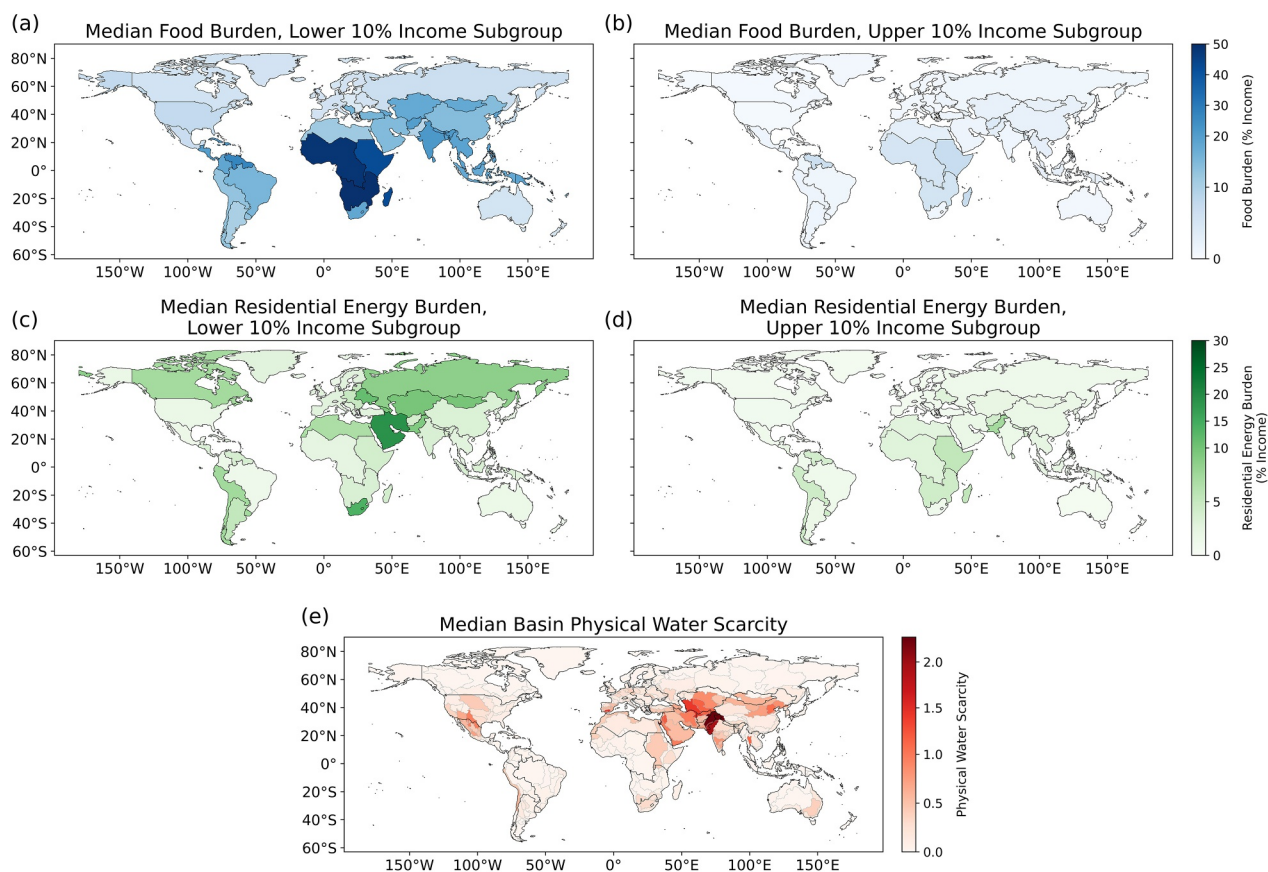


Figure 1. Median resource security outcomes in 2050. (a) Food burden for the lower 10% income subgroup, (b) the upper 10% income subgroup; (c) residential energy burden for the lower 10% income subgroup, (d) the upper 10% income subgroup; and (e) basin-level physical water scarcity. Darker regions and basins indicate regions with lower food, energy, and water security outcomes. Note that the scales of colorbars are different for the three outcomes. In (e), water scarcity is not shown by income subgroups because the Global Change Analysis Model does not currently disaggregate water consumption by household income groups.

Specifically, we use the regression trees to quantify the relative importance of each uncertain factor to regional security outcomes and adopt classification trees to identify the combinations of drivers leading to the scenarios of interest. For each factor's relative importance, we average each factor's reduction in variance (node purity) across 200 regression trees. Classification trees require a subset of scenarios to be classified based on a preset threshold, and the scenarios crossing the threshold are further scrutinized. In our study, we set the thresholds to the 20th percentile (for low burden scenarios) and 80th percentile (for high burden scenarios) indicator for Eastern Europe and Pakistan. For Northern South America, we use the 30th and 70th percentile indicators as our thresholds in order to ensure a sufficient number of scenarios fall into each classification for meaningful pattern discovery. We then fit classification trees to identify the combination of uncertain factors leading to the scenarios below (above) the 20th or 30th (80th or 70th) percentiles.

3. Results

3.1. Future Resource Security Outcomes Are Heterogeneous Across Income Subgroups and Regions

We first explore resource security outcomes across regions, sectors, and income subgroups by highlighting the ensemble median outcomes within these categories in 2050 (Figure 1). Note that we only differentiate income subgroups for food and energy security outcomes, and not for water as shown in Figure 1. Our results suggest that median security outcomes are highly heterogeneous across regions, sectors, and income subgroups. Food and energy burdens differ significantly between the lowest and the highest income subgroups, even in regions with the most extreme outcomes (e.g., sub-Saharan Africa regions for food burden, Middle East for residential energy burden). For instance, Southern Africa's median food burden is a factor of 9 lower in the highest income subgroup compared to the lowest (i.e., 5.5%–49.6%), and the Middle East's median residential energy burden is a factor of

11 lower (i.e., 1.7%–18.9%). Within the lowest income subgroups, the regions facing the greatest food burdens differ substantially from those facing the highest energy burdens. Other regions emerge as hotspots for physical water scarcity. These results demonstrate the importance of investigating outcomes at different scales as they can vary substantially.

In terms of median food burden, by 2050, the population in the lowest income subgroup in sub-Saharan Africa and India is projected to experience the highest food burden outcomes (Figure 1a). Food burdens for the lowest income subgroups in Western, Eastern, and Southern Africa are nearly 50% (48.4%, 42.5%, and 49.6%, respectively) in 2050, greatly surpassing the global median food burden in 2050 of 15.0%. These results are broadly consistent with prior studies highlighting persistent food insecurity in regions with lower income and high population growths (Hasegawa et al., 2015). In addition, increased conservation of natural land from decreasing carbon scenarios and land conservation assumptions also result in an overall increase in staple and nonstaple food prices (Figure S3 in Supporting Information S1). In GCAM, land is a limited resource that multiple sectors compete for, and it is allocated to diverse uses based on expected profit (e.g., cropland, pastureland, forest, etc.) (Wise et al., 2015). In our ensemble, both decreasing carbon and land conservation scenarios reduce the expansion of cropland and pastureland into natural lands. In addition, scenarios with higher bioenergy production and afforestation also increase land competition (Fuhrman et al., 2020). This result indicates that scenarios with large-scale land conservation increase the competition for land, which ultimately lead to surges in crop prices when all other assumptions are held constant. This global increase in crop prices ultimately reduces the food security of sub-populations with lower income levels. This finding aligns with prior research showing that reductions in carbon intensity can increase global food prices and exacerbate food insecurity among vulnerable populations (Obersteiner et al., 2016; Van Meijl, Shutes, et al., 2020). Furthermore, our results demonstrate the importance of a multisectoral modeling approach, especially when modeling future resource security outcomes.

Median values of residential energy burden for the lowest income subgroup are the highest in the Middle East, South Africa, and Eastern Europe, reaching 18.9%, 13.7%, and 11.2%, respectively (Figure 1c). Differently from food demand, which monotonically increases with income in all regions (Figure S4 in Supporting Information S1), demand for residential energy services is more complicated and is affected by multiple factors: income, fuel composition (e.g., electricity, natural gas, biomass, coal), building floorspace, and regional temperature (e.g., heating and cooling degree days) (Sampedro et al., 2024). In particular, we find regions with high demand for residential cooling (e.g., the Middle East) and regions with high reliance on carbon-intensive traditional fuels such as coal and traditional biomass (e.g., South Africa) face the highest residential energy burdens in the lowest income subgroup. In GCAM, traditional fuels are endogenously modeled along with other fuels, and their costs are determined through market equilibrium and technology-specific cost calculations. Therefore, the use of traditional fuels contributes to household energy expenditures and ultimately to residential energy burden. These patterns are consistent with prior studies showing that residential energy burden disproportionately affects low-income households and can be exacerbated under decreasing carbon scenarios that increase energy prices (Hernández, 2016). Our findings imply that drivers leading to high residential energy burdens across regions are highly heterogeneous. The contrast between the Middle East and South Africa motivates a comprehensive analysis across all regions and drivers to derive a robust insight on residential energy burden, which we explore in the following section.

Median water security outcomes (measured in terms of physical water scarcity) are the worst in basins in Pakistan, India, the Middle East, and Central Asia (e.g., Sinai Peninsula, Indus, Sabarmati, Caspian Sea-East Coast, etc.) (Figure 1e). Previous studies have identified that rapid population growth, high degrees of aquifer-reliance, and extensive non-renewable groundwater extraction drive water insecurity in these basins, especially in the first half of the 21st century (Niazi et al., 2024). Water sector outcomes from our scenarios extend the key findings from previous literature (Hejazi et al., 2014) and provide robust results regarding the basins that will face water security challenges in 2050.

3.2. Lower Income Populations Face Higher Uncertainties in Future Resource Security Outcomes

Despite various regional and sectoral dynamics that can be highlighted from the ensemble median, a summary statistic can overlook important features of the ensemble as a whole. For instance, analysts are oftentimes more interested in the uncertainty across scenarios (e.g., overall dispersion of simulations) or in scenarios with extreme outcomes (e.g., scenarios with outcomes greater than a specific percentile) in each region. These outcomes

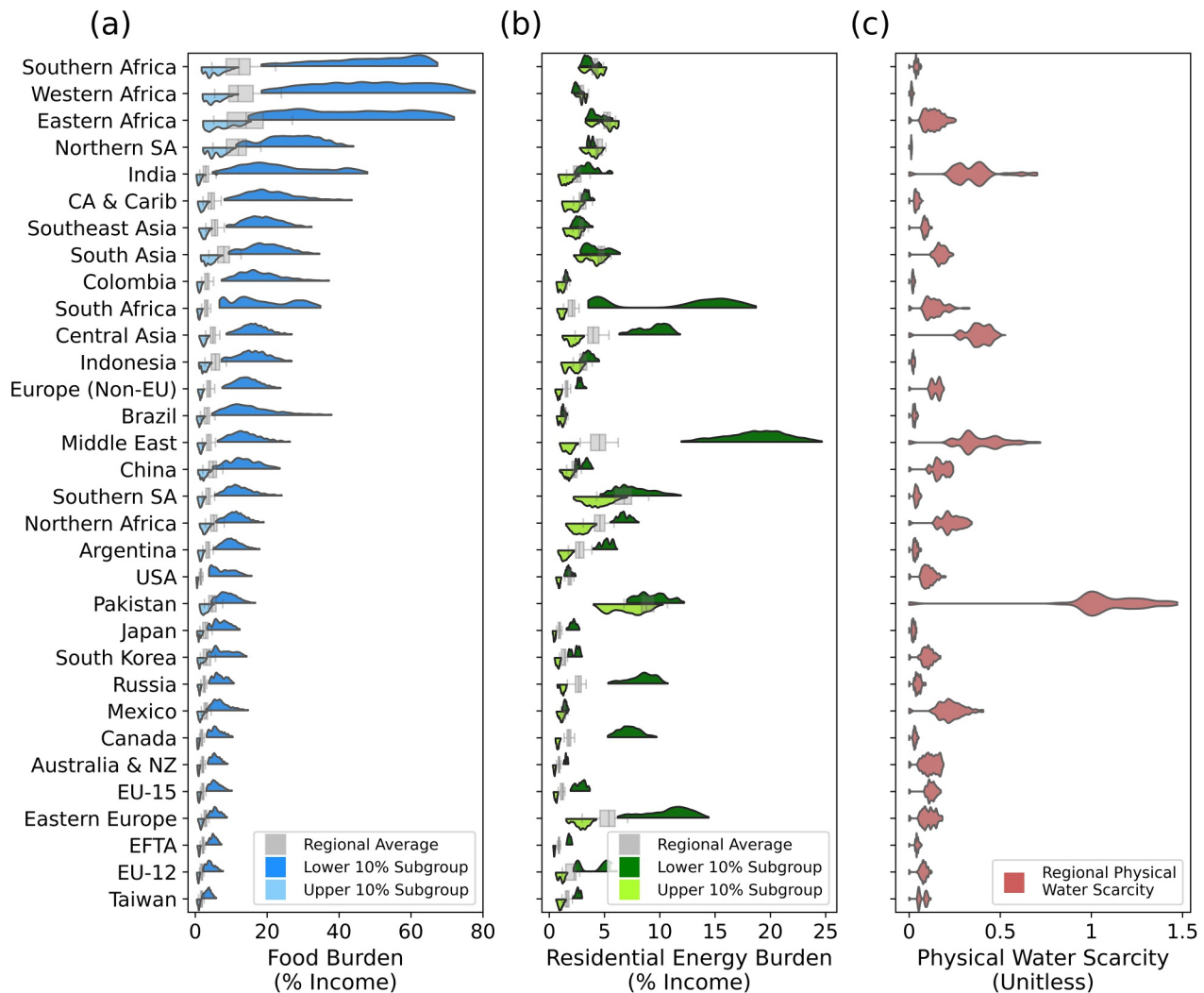


Figure 2. Ranges of resource security outcomes across regions, sectors, and income subgroups. We show the distributions of the outcomes using asymmetric violin plots of (a) food burden, (b) residential energy burden, and (c) physical water scarcity in 2050 across the ensemble. In (a) and (b), the asymmetric violins show the distributions for the lower 10% income subgroup (dark blue and dark green) and the upper 10% income subgroup (light blue and light green) on opposite sides, with boxplots for the regional average (gray). Because we only differentiate across income subgroups for food and energy, we only show the distributions of regional physical water scarcity outcomes in (c). Regions are sorted based on the divergence in food burden between income subgroups, such that regions with the largest differences in food burden appear at the top of the figure. Full names of abbreviated regions are listed in Table S1 in Supporting Information S1.

indicate how the region or sub-population in a region can be impacted by specific scenario assumptions, which can be further examined through scenario discovery techniques.

To explore the uncertainty across simulation outcomes, we visualize the ranges of outcomes across regions, sectors, and income subgroups through asymmetric violin plots (Figure 2). Notably, our results show that the variability across the ensemble is much higher for the lowest income subgroup than for the highest income subgroup, with the range for the lowest income subgroup often falling outside of the range for the regional average (Figures 2a and 2b). This is particularly true in regions where the lowest income subgroup has high burdens (e.g., sub-Saharan Africa and India for food burden; Middle East and South Africa for residential energy burden). For example, the range of possible food burden outcomes for Western Africa's lowest income subgroup spans from 18.5% to 77.5% in 2050 depending on scenario assumptions. Similarly, the residential energy burden for the lowest income subgroup in the Middle East is as low as 12.0% or as high as 24.7%. The high uncertainty across simulation outcomes in lower income subgroups ultimately poses additional pressure on regional food, energy, and water security challenges, especially when deriving robust insights for decision support. Conversely, higher income subgroups are less sensitive to the effects of the uncertain external factors in the ensemble than the lower

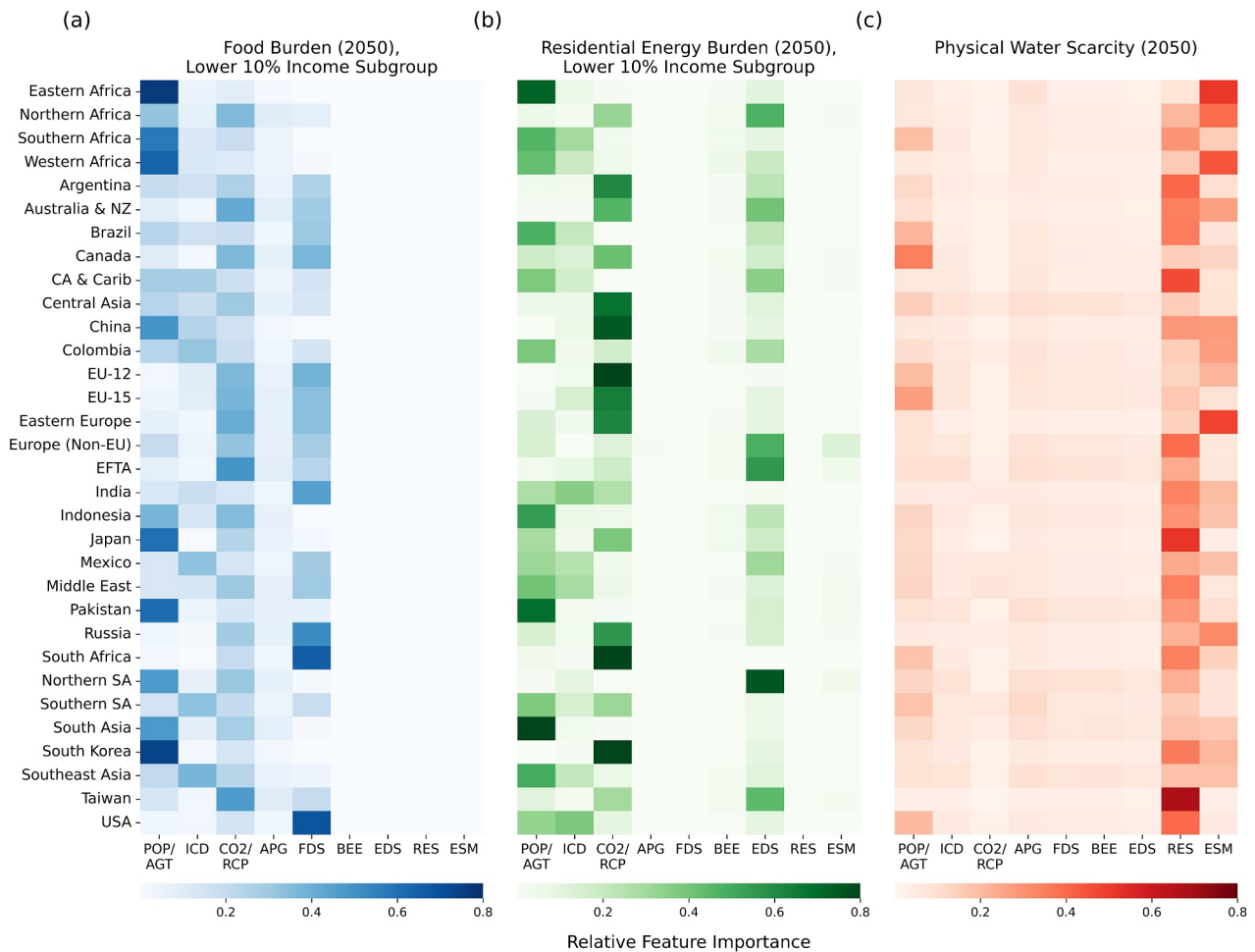


Figure 3. Heatmaps of relative feature importance in 2050 of each factor for three outcomes computed from regression trees: (a) lower 10% income subgroup's food burden, (b) lower 10% income subgroup's residential energy burden, and (c) regional physical water scarcity. Darker values indicate higher relative importance, measured by the increase in node purity. The horizontal axis includes all factors varied in our ensemble design: population, gross domestic product, and level of agricultural trade (POP/AGT), income distributions (ICD), carbon intensity, land conservations, and radiative forcing levels (CO₂/RCP), agricultural productivity growth rate (APG), food demand sensitivity (FDS), building energy efficiency (BEE), energy demand sensitivity (EDS), reservoir expansion (RES), and earth system models (ESM). See Figure S5 in Supporting Information S1 for annotated heatmaps and Figures S6 and S7 in Supporting Information S1 for regression tree results for food and residential energy burden of subgroups with higher income.

income subgroups. This highly differential outcome across income subgroups underscores the value of representing sub-populations when modeling future resource security outcomes, and the dynamics that can be missed by focusing on regional averages.

3.3. Identifying the Key Drivers Influencing Future Resource Security Outcomes

To identify the relative importance of the factors impacting regional resource security outcomes for the lower 10% income subgroup, we employ regression trees and compute each factor's relative feature importance (see Experimental Procedures for details). For food burden, we find that most regions are influenced by socioeconomics (i.e., population, GDP, and income distributions), carbon intensity and land conservation scenarios, and demand sensitivity assumptions acting either singly or in some combination (Figure 3a). Specifically for food burden, we observe that sub-Saharan Africa regions with the highest food burden outcomes are strongly affected by the assumptions on income. This result is specifically driven by a relatively rapid food demand surge in lower income ranges as incomes rise (Figure S4 in Supporting Information S1). In contrast, we find that in high-income economies (e.g., European regions and the USA), food burden is driven primarily by the food demand sensitivity assumptions, which affect consumption behavior. This regional heterogeneity of drivers on food burden outcomes

emphasizes the value of the exploratory modeling approach when exploring resource security outcomes across regions.

From the heatmaps for residential energy burden, we find that the regional key drivers affecting regional residential energy burdens are also socioeconomics, carbon intensity and land conservation, and demand sensitivity assumptions (Figure 3b). Furthermore, we also find that the carbon intensity and land conservation assumptions strongly affect the residential energy burden in South Africa, EU-12, and China. This characteristic results largely from each region's residential energy service composition (e.g., coal, traditional biomass, and modern services) in heating and non-thermal services. Because the population in the lower income subgroups have lower access to modern services, they rely more on carbon-intensive fuels, especially coal (Figures S8–S12 in Supporting Information S1). Consequently, the residential energy burdens of regions with high shares of coal services are highly sensitive to the carbon intensity and land conservation scenarios, in which the costs of carbon-intensive energy sources may be higher.

Finally, for physical water scarcity, we find that the assumptions affecting water supply (i.e., reservoir expansion assumption and hydrologic projections from ESMs) and the factors affecting water demand (i.e., socioeconomics, carbon intensity, and land conservation) are both important (Figure 3c). This implies that in addition to the reservoir expansion assumption, hydrological variability embedded in future hydrologic projections and demand variability in the socioeconomic and carbon intensity assumptions can influence future water scarcity across regions differently. Ultimately, the heterogeneity of influential drivers across regions and sectors motivates identifying the drivers behind joint security outcomes, which we perform a thorough scenario discovery in the following section.

3.4. Regional Income Drives the Relationship Between Joint Food-Energy Security Outcomes

The relationship between joint security outcomes (e.g., joint food-residential energy burden) provides a more holistic picture of a sub-population's resource security. For instance, regions with synergistic relationships between burdens (e.g., both burdens are either high or low together) imply that the relationship between regional food and energy security is relatively straightforward. In contrast, regions with tradeoffs (e.g., one burden is high, and the other is low) indicate that the relationship between different types of resource security can be more complicated. For further analysis, we classify and highlight the relationships between food and residential burdens for the lower 10% income subgroup by mapping the two burdens jointly (Figure 4).

We first note that the relationship between food and residential energy burdens is heterogeneous across regions for lower 10% income subgroup (e.g., synergy, both possibilities, or tradeoff), and income predominantly determines the relationship between burdens. Globally, high-income economies often demonstrate synergy (e.g., European countries, Australia and New Zealand, Japan, Canada, etc.), which is consistent with the expectation from Engel Curves: that at higher incomes, increasing income leads to declines in both food and energy burdens. In some cases, high income countries demonstrate both possibilities (e.g., the USA and Europe Non-EU) between burdens, in which cases other factors varied in the ensemble likely complicate the relationship as incomes rise (Figures S16 and S17 in Supporting Information S1).

However, we observe different dynamics in low-income and lower-middle-income economies, which mainly show tradeoffs between burdens (e.g., regions in Sub-Saharan Africa, Southeast Asia, and South America). Whereas the relationship between food burden and income in these regions is similar to those from high-income economies, residential energy expenditure increases faster than income (Figures S16 and S17 in Supporting Information S1). As previously highlighted, residential energy demand is more complicated than food demand and is influenced by a variety of factors including regional and sub-population income, regional building floorspace, temperature impacts on energy demand, and fuel composition (L. Clarke et al., 2018; Sampedro et al., 2022a). These factors, individually or combined, contribute to rapid increases in residential energy expenditures in different ways across regions. Our results generally highlight that regions with relatively low incomes that are located near the equator will ultimately result in a tradeoff between burden outcomes. The synergistic relationship in high-income economies and trade-off relationship in low-income economies are further supported by the shapes of Engel curves which, as previously discussed, change based on income levels (Dorband et al., 2019). Within each region, the range of GDP also varies. For example, synergistic regions (e.g., EFTA, Japan, and South Korea) tend to appear at the higher end of the income scale (dark blue), whereas tradeoff regions (e.g., Mexico, Indonesia, and Brazil) show a wider spread of income levels across scenarios (mix of blue and red).

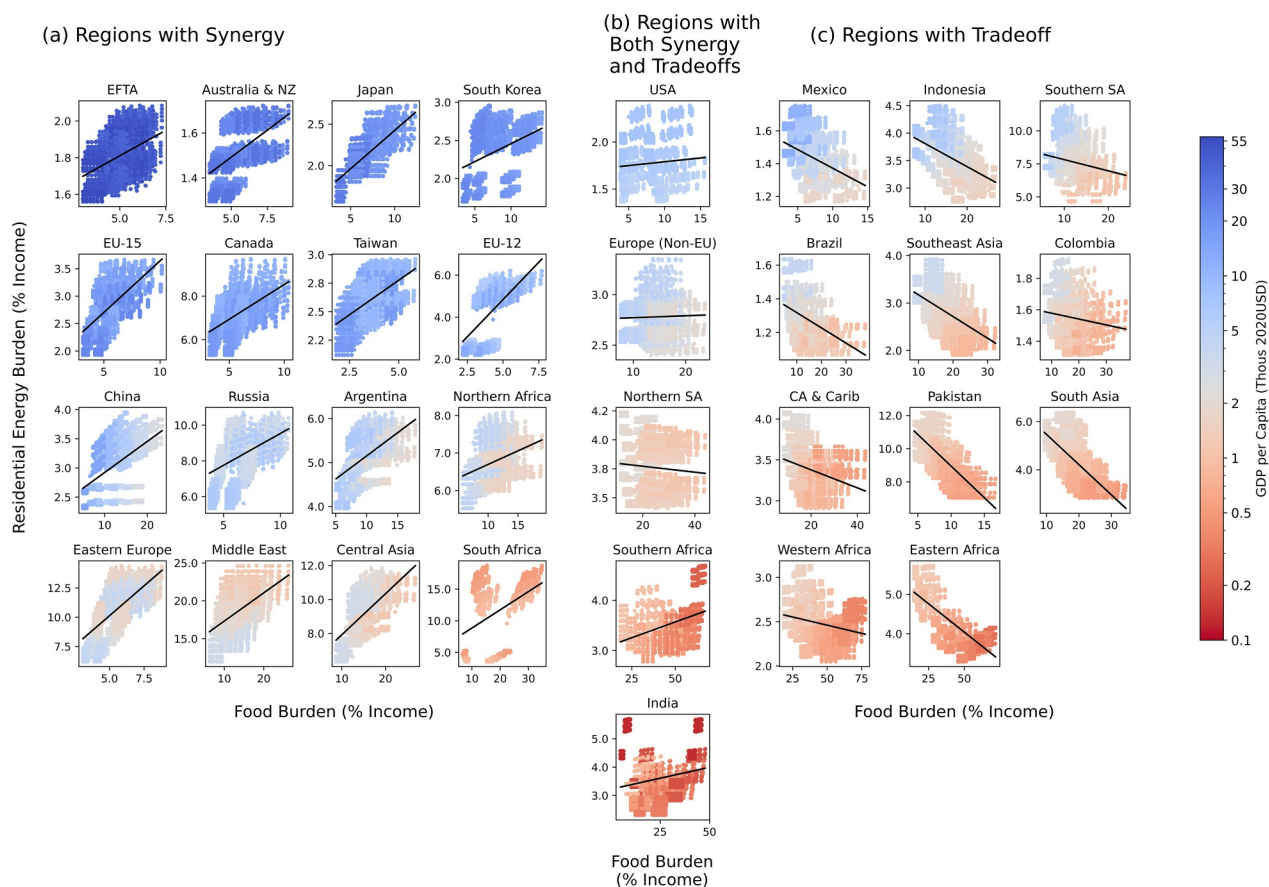


Figure 4. Scatterplots and the line of best fit of joint food (x-axis) and residential energy (y-axis) burdens across scenarios for the lower 10% income subgroup in 2050, classified by their relationships: (a) regions with synergy, (b) regions with both synergy and tradeoff, and (c) regions with tradeoff. Points are colored by sub-population income (thousands of 2020 USD), with the middle color corresponding to the median income of this subgroup across regions. The colors represent the sub-population's income corresponding to each scenario. For joint food-residential energy burden outcomes for higher income subgroups and with base year (2015) values added to the plot, see Figures S13–S15 in Supporting Information S1.

Notably, our results also show that regions with the lowest income levels such as Southern Africa and India are classified as regions with both possible relationships, not with just tradeoffs. This implies that while income is a prevailing factor, other factors and local characteristics further determine the relationship between burden outcomes. This motivates us to perform scenario discovery to identify the combination of drivers leading to extreme scenarios in regions with different relationships between burdens.

3.5. Combination of Drivers Lead to Extreme Joint Resource Security Outcomes

Scenario discovery enables analysts to discover the pathways leading to outcomes of interest, which can ultimately inform local policy decisions when designing long-term strategies to enhance regional resource security. To identify the combinations of drivers leading to the outcomes of interest, we further divide selected regions' joint security outcomes into four subsets of scenarios: (a) jointly low, (b) jointly high, (c) high food, and (d) high energy burden scenarios. We then apply scenario discovery to each subset of scenarios in order to identify the combination of drivers leading to the subset. For scenario discovery, we select three regions with (a) the strongest synergy (Eastern Europe), (b) the strongest tradeoff (Pakistan), and (c) both possibilities (Northern South America) between joint burden outcomes (Figure S18 in Supporting Information S1).

Nearly 50% of jointly low burden scenarios in Eastern Europe (i.e., 42.7%; 216 of 506 scenarios) (Figure 5a) share the following assumptions: (a) increasing carbon scenarios, (b) high income (SSP5 population and GDP assumptions), and (c) more sensitive consumer behavior to price (high food demand sensitivity). This implies that jointly low outcome scenarios are driven by all factors affecting the burden indicator: (a) lower prices of

Eastern Europe (Synergy), Lower 10% Income Subgroup

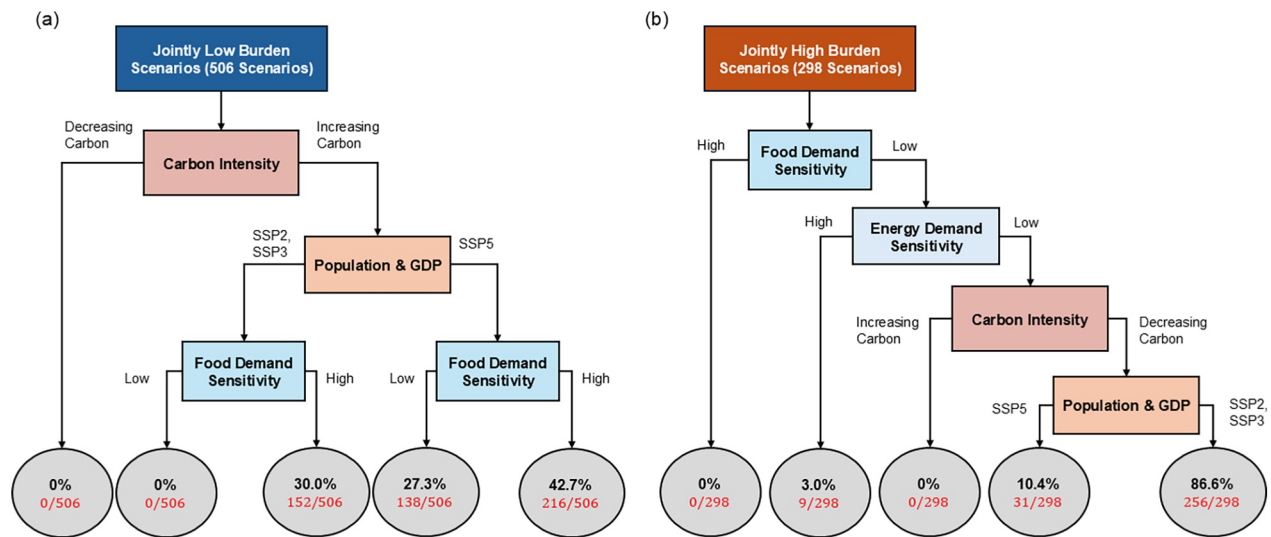


Figure 5. Simplified classification trees for Eastern Europe's joint food-residential energy burden outcomes for the lower 10% income subgroup. Classification trees show the combination of drivers leading to (a) jointly low and (b) jointly high burden scenarios. Percentage in each node is coverage, which represents the percentage of scenarios following the pathway (may not add up exactly to 100% due to rounding). Full classification trees for Eastern Europe are included in Figure S19 in Supporting Information S1.

residential energy and food, (b) higher income, and (c) if prices do increase, then decreased demand for food and residential energy services, limiting expenditures. On the other hand, 86.6% of jointly high burden scenarios in Eastern Europe (Figure 5b) are primarily driven by (a) less sensitive consumer behavior to price (lower food and energy demand sensitivities), (b) decreasing carbon scenarios, and (c) medium or low income (SSP2 or SSP3 population and GDP assumptions). This combination of drivers results in scenarios with higher prices, lower income, and relatively price inelastic demand, increasing food and residential energy expenditures. The combination of drivers identified from classification trees demonstrates how factors other than income (e.g., carbon intensity and demand sensitivity assumptions) shape jointly low and high burden scenarios in synergistic regions.

However, classification trees for Pakistan's security outcomes (Figure 6) illustrate different results compared to those from Eastern Europe. Because Pakistan's food and residential energy burdens show a strong tradeoff relationship, we highlight the combination of drivers toward either high food or high residential energy burden scenarios. The pathways toward Pakistan's high food burden scenarios include (a) low income (SSP3 population, GDP, and income distribution assumptions), (b) less sensitive consumer behavior to price (lower food demand sensitivity), which makes food expenditures increase more substantially with higher prices, and (c) lower agricultural productivity (low or medium agricultural productivity growth rates), which generally implies higher food prices (Figure 6a). The importance of the agricultural productivity growth rate assumption is influenced by the basins with high crop production in Pakistan (e.g., Sabarmati and Indus basins), which implies that the pathways toward high burden outcomes can be highly localized depending on regional characteristics. On the other hand, high residential energy burden scenarios for Pakistan include (a) high income (SSP5 population and GDP assumptions), which moves the subgroup's income into an income range in which residential energy expenditure rises faster than income, (b) less sensitive consumer behavior to price (reference energy demand sensitivity) so that expenditures increase with higher prices, and (c) increasing carbon scenarios, in which food prices do not increase as much, limiting the food burden. Differently from Eastern Europe's driver toward high burden outcomes, Pakistan includes high income as the key driver to high residential energy burden scenarios. This is because the regions with tradeoffs between burdens tend to have a large gap between actual energy demand and consumption level that would satiate demand (i.e., a large satiation gap). Therefore, under scenarios with high income, consumers increase their residential energy extensively to close this gap, which leads to higher residential energy expenditures. As a result, these scenarios show low food burden but high residential energy burden, ultimately characterizing them as regions with tradeoffs between security outcomes.

Pakistan (Tradeoff), Lower 10% Income Subgroup

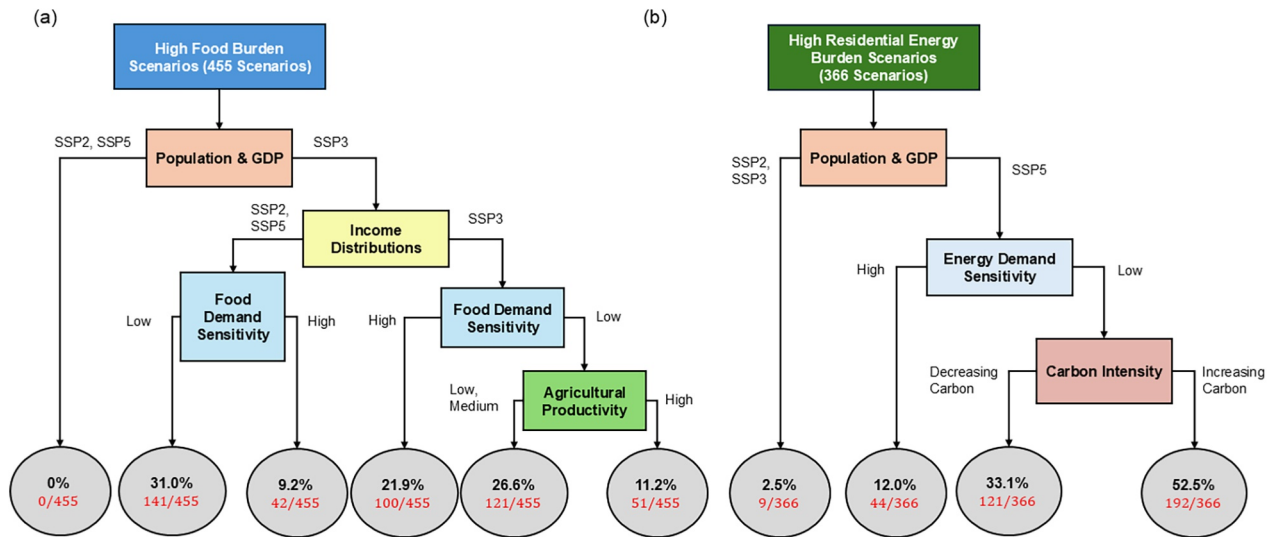


Figure 6. Simplified classification trees for Pakistan's joint food-residential energy burden outcomes for the lower 10% income subgroup. Classification trees show the combination of drivers leading to (a) high food and (b) high residential energy burden scenarios. Full classification trees for Pakistan are included in Figure S20 in Supporting Information S1.

Finally, we highlight a region with both possibilities (i.e., Northern South America) and identify the combination of drivers leading to high residential energy (Figure 7a) and jointly low burden (Figure 7b) scenarios. We first find that Northern South America's combination of drivers toward high residential energy burden scenarios is analogous to Pakistan's: (a) less sensitive consumer behavior to price (lower energy demand sensitivity), (b) high income (SSP5 population and GDP assumptions), and (c) increasing carbon scenarios or decreasing carbon scenarios without land conservation. This implies that residential energy demand expenditures increase faster than income in a subset of scenarios in Northern South America, resulting in scenarios with high residential

Northern South America (Both Tradeoff and Synergy), Lower 10% Income Subgroup

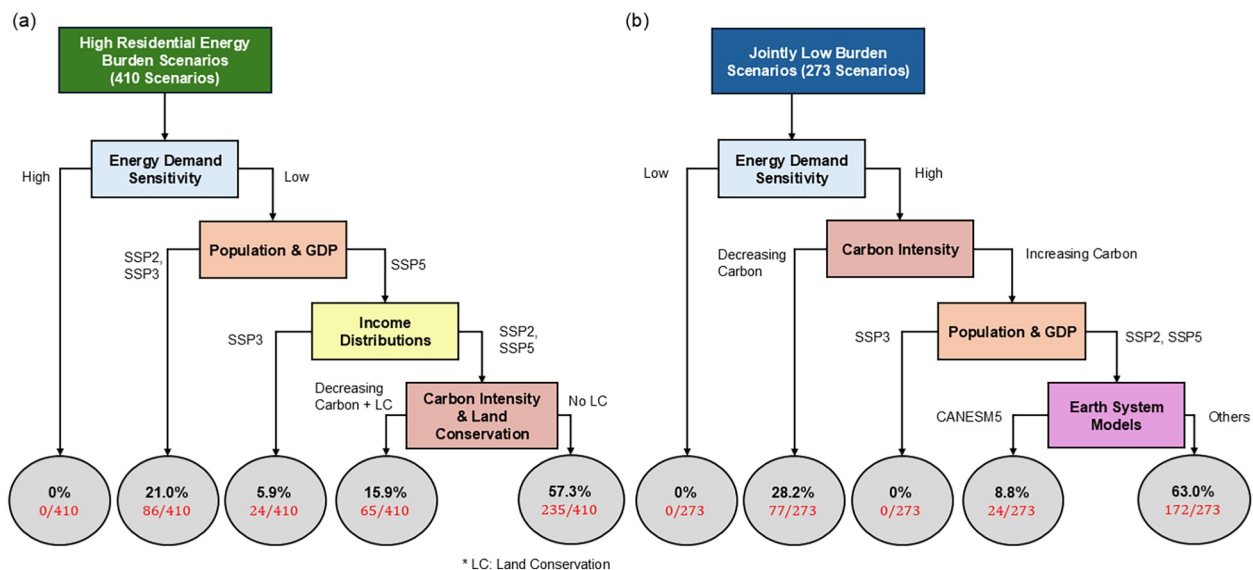


Figure 7. Simplified classification trees for Northern South America's joint food-residential energy burden outcomes for the lower 10% income subgroup. Classification trees show the combination of drivers leading to (a) high residential energy and (b) jointly low burden scenarios. Full classification trees for Northern South America are included in Figure S21 in Supporting Information S1.

energy burdens. At the same time, we also observe an opposite relationship between residential energy demand expenditures and income in a different subset of scenarios. Under these scenarios, income increases faster than residential energy expenditure, resulting in scenarios with jointly low burden outcomes. Here, Northern South America's combination of drivers toward jointly low burden outcomes overlaps with Eastern Europe's drivers: (a) more sensitive consumer behavior to price, (b) lower food and energy prices, and (c) high income. However, scenarios with jointly low burdens in Northern South America do not necessarily indicate equivalent overall resource security to Eastern Europe's. Whereas Eastern Europe's low burden scenarios are mainly from high income and lower prices, Northern South America's outcomes are due to lower residential energy demand, resulting in a large satiation gap in residential energy. As illustrated through jointly low burden scenarios in Northern South America, our burden indicators do not capture a full range of resource security dimensions, such as the ability to meet residential energy satiation levels. We plan to further explore overall security outcomes in our scenario ensemble through additional indicators in future studies.

Based on the results highlighted from the classification trees in selected regions (Figures 5–7), we can draw several general insights: (a) income predominantly determines the relationship between burden outcomes (i.e., synergy, both possibilities, and tradeoff, see also Figures S13 and S14 in Supporting Information S1), (b) non-income factors (e.g., demand sensitivities and carbon intensity and land conservation scenarios) and regions' local characteristics (e.g., geographical location, fuel compositions, land use, etc.) also affect the relationship between burdens, and (c) the combination of drivers that lead to low and high burden scenarios jointly and separately are heterogeneous across regions with synergy, tradeoff, and both possibilities. These insights from joint burden outcomes and classification tree analyses can ultimately contribute to the formulation of robust policy decisions to enhance future resource security.

4. Conclusions

Resource security refers to reliable and sustained access to essential resources such as food, energy, and water. Until now, various research communities have utilized existing community scenario frameworks (e.g., the SSP-RCP scenarios) to facilitate research across a wide range of topics on resource security. However, the factors that comprise these scenario frameworks are broadly designed and lack details to directly answer research questions on the determinants of resource security outcomes. Moreover, no study has systematically explored the impacts of future uncertain factors on security outcomes of multiple sub-populations defined by income levels. Therefore, it is important to develop scenarios specifically focused on examining future resource security outcomes across multiple income subgroups. To address this research gap, we generate a large ensemble of global change scenarios by varying the factors that directly affect future resource security outcomes, including socioeconomics, carbon intensity and land conservation scenarios, food, energy, water, and temperature change impacts. Our ensemble covers a wide range of plausible futures at sub-regional, regional, and global scales, including differentiated security outcomes across 10 income subgroups within each region. Using the ensemble, we explore how various uncertain factors shape future resource security outcomes across regions, sectors, and income subgroups. Furthermore, we adopt scenario discovery to identify the combination of drivers that lead to extreme outcomes, both separately and jointly, across regions with different relationships between joint outcomes.

Our key results first highlight the heterogeneous impacts of future uncertainties on security outcomes across sectors, regions, and income subgroups in 2050. For instance, the sub-populations with the lowest income in sub-Saharan Africa and India experience the highest food burden outcomes of nearly 50%. This is due to low regional income and high food prices that arise in scenarios with decreasing carbon and land conservation. On the other hand, the Middle East and South Africa emerge as regions with the highest residential energy burden for the lowest income group for different reasons: high demand for residential cooling and high shares of traditional fuels. Water scarcity challenges exist in South Asia, the Middle East, and Northern Africa due to rapid demand growth and extensive groundwater extraction. Across all sectors, we demonstrate that the regional average outcomes obscure both extremely high burden outcomes for lower-income subgroups and extremely low burden outcomes for higher-income subgroups. For instance, the range of food burden for Western Africa's lowest income population extends from 18.5% to 77.5%, while the regional average only ranges between 5.4% and 24.0%. This emphasizes the importance of modeling sub-regional populations when considering resource security outcomes, which would be lost if only regional average outcomes were considered. Furthermore, the heterogeneous outcomes and impacts across sectors, regions, and income subgroups underscore the value of the multisectoral, exploratory modeling approach we adopt to model future resource security outcomes.

We further identify the key drivers of security outcomes across regions and quantify each driver's relative contribution. In general, we find that socioeconomics (i.e., GDP, population, and income distributions), carbon intensity and land conservation scenarios, and consumer behavior assumptions heavily affect food and energy sector outcomes for the lowest income subgroup. Notably, scenarios with lower carbon may have an increased energy burden for the low-income population reliant on traditional carbon-intensive fuels (e.g., coal). Physical water scarcity outcomes are most strongly influenced by the reservoir expansion assumption across all regions. After that, uncertain factors affecting water supply (e.g., global runoff projections) and water demand (e.g., socioeconomics, carbon intensity and land conservation scenarios) affect the outcomes, depending on the region. Our results are consistent with other studies on future water scarcity considering global change (Birnbbaum et al., 2022; Leijnse et al., 2024).

We then classify regions into three categories based on the relationship between joint food-residential energy burden outcomes (i.e., regions with synergies, tradeoffs, and both possibilities) and perform scenario discovery on selected regions. While the joint relationship between burdens across many regions is predominantly determined by income level, we discover the impacts from non-income factors by fitting classification trees to extreme scenarios. For instance, classification trees for the region with the strongest synergy (Eastern Europe) identify that scenarios with low income coupled with decreasing carbon and less sensitive consumer behavior to prices result in jointly high burden outcomes. In contrast, scenarios with high income results in high residential energy burden outcomes in the lower-middle-income economy with strong tradeoff (Pakistan). While food demand is relatively inelastic to income across all regions, residential energy demand is more elastic to income, especially in lower- and lower-middle income economies with high difference between actual demand and consumption level that would satiate demand. In these regions, an increase in income reduces food burden but simultaneously elevates residential energy burden due to increased demand for energy services approaching satiation levels. In the region with both synergy and tradeoff (Northern South America), scenarios experience both an increase and decrease in residential energy expenditures with the change in income. We also discover that regional key drivers are dependent on local characteristics, such as land use and temperature change impacts on water and energy sectors. Our scenario discovery on joint food-residential energy burden outcomes across regions illustrates how intersecting drivers shape regional resource security outcomes.

Although a number of insights on resource security outcomes are identified through a large ensemble of scenarios, the modeling and analysis contained in this study could be further developed by (a) expanding the temporal scope of the analysis, (b) including a wider range of human well-being outcome indicators, and (c) conducting a model intercomparison exercise using a subset of scenarios from the ensemble to enable broader community insights and to support future scenario development. First, our analysis focuses on resource security outcomes in 2050, but these outcomes change over time. Analyzing the trajectories of security outcomes evolving over the 21st century can reveal how they improve, worsen, or shift across regions and income subgroups. Second, we present resource security outcomes using food burden, residential energy burden, and physical water scarcity. While our analysis provides a starting point for scenario generation by incorporating essential dimensions of resource security, it is important to note that these indicators do not comprehensively capture all aspects of resource security. The use of burden indicators as a proxy offers a tractable approach to explore economic access to food and energy but they do not directly measure consumption, nutrition, or adaptive behaviors. Future research should aim to integrate multi-dimensional indicators of resource security (and, more broadly, human well-being), including income or wealth inequality, human health, food consumption relative to minimum dietary standards, dietary diversity, and satiation gap in heating and cooling services. Doing so would enable a more holistic analysis of human well-being across sectors and populations. Until such integration, our analysis should be interpreted primarily as a methodological demonstration of the multisector modeling framework rather than a holistic assessment of resource security at household-level.

Finally, conducting a model intercomparison exercise using a subset of scenarios from the ensemble enables the broader community to glean insights for further model development. Based on the insights from scenario discovery, one can choose which inputs to include when generating a different set of scenarios for further analysis. Ultimately, we expect our ensemble and insights to potentially inform producing new sets of community scenarios (e.g., community scenarios following the RCP-SSP scenario framework).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

GCAM is an open-source model available at <https://github.com/JGCRI/gcam-core>. The code to run the ensemble, query output databases, process query output data, and generate all figures is published on Zenodo at Kim (2025).

Acknowledgments

This research was supported by the U.S. Department of Energy, Office of Science, as part of research in MultiSector Dynamics, Earth and Environmental System Modeling Program. A portion of the effort by Gi Joo Kim was also supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (RS-2024-00405470). A portion of the effort by Jennifer Morris was also supported by the U.S. Department of Energy, Office of Science, under Award Number DE-FG02-94ER61937. A portion of the effort by John Weyant was also supported by Stanford University internal funding. The Pacific Northwest National Laboratory is operated for DOE by Battelle Memorial Institute under contract DE-AC05-76RL01830. The views and opinions expressed in this paper are those of the authors alone.

References

- Bankes, S. (1993). Exploratory modeling for policy analysis. *Operations Research*, 41(3), 435–449. <https://doi.org/10.1287/opre.41.3.435>
- Bauen, A. (2006). Future energy sources and systems—Acting on climate change and energy security. *Journal of Power Sources*, 157(2), 893–901. <https://doi.org/10.1016/j.jpowsour.2006.03.034>
- Birnbaum, A., Lamontagne, J., Wild, T., Dolan, F., & Yarlagadda, B. (2022). Drivers of future physical water scarcity and its economic impacts in Latin America and the Caribbean. *Earth's Future*, 10(8), e2022EF002764. <https://doi.org/10.1029/2022EF002764>
- Bond-Lamberty, B., Patel, P., Lurz, J., Kyle, P., kvcalvin, Smith, S., et al. (2024). JGCRI/gcam-core: GCAM 7.1 (version gcam-v7.1). *Zenodo*. <https://doi.org/10.5281/ZENODO.11481167>
- Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (2017). In *Classification and regression trees* (1st ed.). Routledge. <https://doi.org/10.1201/9781315139470>
- Bryant, B. P., & Lempert, R. J. (2010). Thinking inside the box: A participatory, computer-assisted approach to scenario discovery. *Technological Forecasting and Social Change*, 77(1), 34–49. <https://doi.org/10.1016/j.techfore.2009.08.002>
- Calvin, K., Bond-Lamberty, B., Clarke, L., Edmonds, J., Eom, J., Hartin, C., et al. (2017). The SSP4: A world of deepening inequality. *Global Environmental Change*, 42, 284–296. <https://doi.org/10.1016/j.gloenvcha.2016.06.010>
- Calvin, K., Patel, P., Clarke, L., Asrar, G., Bond-Lamberty, B., Cui, R. Y., et al. (2019). GCAM v5.1: Representing the linkages between energy, water, land, climate, and economic systems. *Geoscientific Model Development*, 12(2), 677–698. <https://doi.org/10.5194/gmd-12-677-2019>
- Cameron, C., Pachauri, S., Rao, N. D., McCollum, D., Rogelj, J., & Riahi, K. (2016). Policy trade-offs between climate mitigation and clean cook-stove access in south Asia. *Nature Energy*, 1(1), 15010. <https://doi.org/10.1038/nenergy.2015.10>
- Clarke, J. F., & Edmonds, J. A. (1993). Modelling energy technologies in a competitive market. *Energy Economics*, 15(2), 123–129. [https://doi.org/10.1016/0140-9883\(93\)90031-L](https://doi.org/10.1016/0140-9883(93)90031-L)
- Clarke, L., Eom, J., Marten, E. H., Horowitz, R., Kyle, P., Link, R., et al. (2018). Effects of long-term climate change on global building energy expenditures. *Energy Economics*, 72, 667–677. <https://doi.org/10.1016/j.eneco.2018.01.003>
- Clements, K. W., & Si, J. (2018). Engel's law, diet diversity, and the quality of food consumption. *American Journal of Agricultural Economics*, 100(1), 1–22. <https://doi.org/10.1093/ajae/aax053>
- Dolan, F., Lamontagne, J., Calvin, K., Snyder, A., Narayan, K. B., Di Vittorio, A. V., & Vernon, C. R. (2022). Modeling the economic and environmental impacts of land scarcity under deep uncertainty. *Earth's Future*, 10(2), e2021EF002466. <https://doi.org/10.1029/2021EF002466>
- Dolan, F., Lamontagne, J., Link, R., Hejazi, M., Reed, P., & Edmonds, J. (2021). Evaluating the economic impact of water scarcity in a changing world. *Nature Communications*, 12(1), 1915. <https://doi.org/10.1038/s41467-021-22194-0>
- Dorband, I. I., Jakob, M., Kalkuhl, M., & Steckel, J. C. (2019). Poverty and distributional effects of carbon pricing in low- and middle-income countries – A global comparative analysis. *World Development*, 115, 246–257. <https://doi.org/10.1016/j.worlddev.2018.11.015>
- Edmonds, J. A., Link, R., Waldhoff, S. T., & Cui, R. (2017). A global food demand model for the assessment of complex human-earth systems. *Climate Change Economics*, 8(4), 1750012. <https://doi.org/10.1142/S2010007817500129>
- FAO. (2023). The state of food security and nutrition in the world 2023. FAO; IFAD; UNICEF; WFP; WHO. <https://doi.org/10.4060/cc3017en>
- Fuhrman, J., McJeon, H., Patel, P., Doney, S. C., Shobe, W. M., & Clarend, A. F. (2020). Food–energy–water implications of negative emissions technologies in a +1.5°C future. *Nature Climate Change*, 10(10), 920–927. <https://doi.org/10.1038/s41558-020-0876-z>
- Gambhir, A., George, M., McJeon, H., Arnell, N. W., Bernie, D., Mittal, S., et al. (2022). Near-term transition and longer-term physical climate risks of greenhouse gas emissions pathways. *Nature Climate Change*, 12(1), 88–96. <https://doi.org/10.1038/s41558-021-01236-x>
- Graham, N. T., Iyer, G., Wild, T. B., Dolan, F., Lamontagne, J., & Calvin, K. (2023). Agricultural market integration preserves future global water resources. *One Earth*, 6(9), 1235–1245. <https://doi.org/10.1016/j.oneear.2023.08.003>
- Hasegawa, T., Fujimori, S., Takahashi, K., & Masui, T. (2015). Scenarios for the risk of hunger in the twenty-first century using shared socioeconomic pathways. *Environmental Research Letters*, 10(1), 014010. <https://doi.org/10.1088/1748-9326/10/1/014010>
- Hejazi, M. I., Edmonds, J., Clarke, L., Kyle, P., Davies, E., Chaturvedi, V., et al. (2014). Integrated assessment of global water scarcity over the 21st century under multiple climate change mitigation policies. *Hydrology and Earth System Sciences*, 18(8), 2859–2883. <https://doi.org/10.5194/hess-18-2859-2014>
- Hernández, D. (2016). Understanding “energy insecurity” and why it matters to health. *Social Science and Medicine*, 167, 1–10. <https://doi.org/10.1016/j.socscimed.2016.08.029>
- Iyer, G., Ou, Y., Edmonds, J., Fawcett, A. A., Hultman, N., McFarland, J., et al. (2022). Ratcheting of climate pledges needed to limit peak global warming. *Nature Climate Change*, 12(12), 1129–1135. <https://doi.org/10.1038/s41558-022-01508-0>
- Kim, G. J. (2025). Model, Data, and Codes for Kim et al. “Identifying the Uncertainties and Drivers of Future Human Outcomes through a Multisector Scenario Ensemble.” [Dataset]. <https://doi.org/10.5281/ZENODO.15587981>
- Kim, G. J., Wessel, J., Birnbaum, A., Moraites, G., Snyder, A., Morris, J., et al. (2025). Discovering the multisectoral impacts of global energy sector outcomes through multiple ensemble aggregation measures. *Earth's Future*, 13(10), e2025EF006526. <https://doi.org/10.1029/2025EF006526>
- Kling, C. L., Arriitt, R. W., Calhoun, G., & Keiser, D. A. (2017). Integrated assessment models of the food, energy, and water nexus: A review and an outline of research needs. *Annual Review of Resource Economics*, 9(1), 143–163. <https://doi.org/10.1146/annurev-resource-100516-033533>
- Lamontagne, J. R., Reed, P. M., Link, R., Calvin, K. V., Clarke, L. E., & Edmonds, J. A. (2018). Large ensemble analytic framework for consequence-driven discovery of climate change scenarios. *Earth's Future*, 6(3), 488–504. <https://doi.org/10.1002/2017EF000701>
- Lange, S., & Büchner, M. (2021). ISIMIP3b bias-adjusted atmospheric climate input data (version 1.1) [Dataset]. *ISIMIP Repository*. <https://doi.org/10.48364/ISIMIP.842396.1>

- Leijnse, M., Bierkens, M. F. P., Gommans, K. H. M., Lin, D., Tait, A., & Wanders, N. (2024). Key drivers and pressures of global water scarcity hotspots. *Environmental Research Letters*, 19(5), 054035. <https://doi.org/10.1088/1748-9326/ad3c54>
- Lele, U., Masters, W. A., Kinabo, J., Meenakshi, J. V., & Ramaswami, B. (2016). Measuring food and nutrition security: An independent technical assessment and user's guide for existing indicators. *Measuring Food and Nutrition Security Technical Working Group. Rome: Food Security Information Network*. <http://www.fsincop.net/topics/fns-measurement>
- Li, X., Vernon, C. R., Hejazi, M. I., Link, R. P., Feng, L., Liu, Y., & Rauchenstein, L. T. (2017). Xanthos – A global hydrologic model. *Journal of Open Research Software*, 5(1), 21. <https://doi.org/10.5334/jors.181>
- Marchau, V. A. W. J., Walker, W. E., Bloemen, P. J. T. M., & Popper, S. W. (Eds.) (2019). *Decision making under deep uncertainty: From theory to practice*. Springer International Publishing. <https://doi.org/10.1007/978-3-030-05252-2>
- Narayan, K., Vernon, C., Waldhoff, S., Edmonds, J., & Cui, R. (2021). Ambrosia: An R package for calculating and analyzing food demand that is responsive to changing incomes and prices. *Journal of Open Source Software*, 6(59), 2890. <https://doi.org/10.21105/joss.02890>
- Narayan, K. B., O'Neill, B. C., Waldhoff, S., & Tebaldi, C. (2024). A consistent dataset for the net income distribution for 190 countries and aggregated to 32 geographical regions from 1958 to 2015. *Earth System Science Data*, 16(5), 2333–2349. <https://doi.org/10.5194/essd-16-2333-2024>
- Narayan, K. B., O'Neill, B. C., Waldhoff, S. T., & Tebaldi, C. (2023). Non-parametric projections of national income distribution consistent with the shared Socioeconomic pathways. *Environmental Research Letters*, 18(4), 044013. <https://doi.org/10.1088/1748-9326/acbdb0>
- Niazi, H., Wild, T. B., Turner, S. W. D., Graham, N. T., Hejazi, M., Msangi, S., et al. (2024). Global peak water limit of future groundwater withdrawals. *Nature Sustainability*, 7(4), 413–422. <https://doi.org/10.1038/s41893-024-01306-w>
- Obersteiner, M., Walsh, B., Frank, S., Havlík, P., Cantele, M., Liu, J., et al. (2016). Assessing the land resource–food price nexus of the sustainable development goals. *Science Advances*, 2(9), e1501499. <https://doi.org/10.1126/sciadv.1501499>
- O'Neill, B. C., Carter, T. R., Ebi, K., Harrison, P. A., Kemp-Benedict, E., Kok, K., et al. (2020). Achievements and needs for the climate change scenario framework. *Nature Climate Change*, 10(12), 1074–1084. <https://doi.org/10.1038/s41558-020-00952-0>
- O'Neill, B. C., Morris, J., Lamontagne, J., Weyant, J., & Wise, M. (2024). A framework for multisector scenarios of outcomes for well-being and resilience. *Earth's Future*, 12(5), e2023EF004343. <https://doi.org/10.1029/2023EF004343>
- Ou, Y., Iyer, G., Clarke, L., Edmonds, J., Fawcett, A. A., Hultman, N., et al. (2021). Can updated climate pledges limit warming well below 2°C? *Science*, 374(6568), 693–695. <https://doi.org/10.1126/science.abl8976>
- Pachauri, S., Van Ruijven, B. J., Nagai, Y., Riahi, K., Van Vuuren, D. P., Brew-Hammond, A., & Nakicenovic, N. (2013). Pathways to achieve universal household access to modern energy by 2030. *Environmental Research Letters*, 8(2), 024015. <https://doi.org/10.1088/1748-9326/8/2/024015>
- Ravenel, R. M., & Redford, K. H. (2009). Understanding IUCN protected area categories. *Natural Areas Journal*, 25(4), 381–389.
- Sachs, J. D. (2012). From millennium development goals to sustainable development goals. *The Lancet*, 379(9832), 2206–2211. [https://doi.org/10.1016/S0140-6736\(12\)60685-0](https://doi.org/10.1016/S0140-6736(12)60685-0)
- Sampedro, J., Iyer, G., Msangi, S., Waldhoff, S., Hejazi, M., & Edmonds, J. A. (2022a). Implications of different income distributions for future residential energy demand in the U.S. *Environmental Research Letters*, 17(1), 014031. <https://doi.org/10.1088/1748-9326/ac43df>
- Sampedro, J., Iyer, G., Msangi, S., Waldhoff, S., Hejazi, M., & Edmonds, J. A. (2022b). Implications of different income distributions for future residential energy demand in the U.S. *Environmental Research Letters*, 17(1), 014031. <https://doi.org/10.1088/1748-9326/ac43df>
- Sampedro, J., Waldhoff, S. T., Edmonds, J. A., Iyer, G., Msangi, S., Narayan, K. B., et al. (2024). Residential energy demand, emissions, and expenditures at regional and income-decile level for alternative futures. *Environmental Research Letters*, 19(8), 084031. <https://doi.org/10.1088/1748-9326/ad6015>
- Stiglitz, P. J. E., Sen, P. A., & Fitoussi, P. J.-P. (2009). Report by the commission on the measurement of economic performance and social progress.
- Tilman, D., & Clark, M. (2014). Global diets link environmental sustainability and human health. *Nature*, 515(7528), 518–522. <https://doi.org/10.1038/nature13959>
- Turner, S. W. D., Hejazi, M., Yonkofski, C., Kim, S. H., & Kyle, P. (2019). Influence of groundwater extraction costs and resource depletion limits on simulated global nonrenewable water withdrawals over the twenty-first century. *Earth's Future*, 7(2), 123–135. <https://doi.org/10.1029/2018EF001105>
- UNDP. (1994). *Human development report 1994*. Oxford Univ. Pr.
- Van Meijl, H., Shutes, L., Valin, H., Stehfest, E., Van Dijk, M., Kuiper, M., et al. (2020b). Modelling alternative futures of global food security: Insights from FOODSECURE. *Global Food Security*, 25, 100358. <https://doi.org/10.1016/j.gfs.2020.100358>
- Van Meijl, H., Tabeau, A., Stehfest, E., Doelman, J., & Lucas, P. (2020a). How food secure are the green, rocky and middle roads: Food security effects in different world development paths. *Environmental Research Communications*, 2(3), 031002. <https://doi.org/10.1088/2515-7620/ab7aba>
- Van Vuuren, D. P., Bijl, D. L., Bogaart, P., Stehfest, E., Biemans, H., Dekker, S. C., et al. (2019). Integrated scenarios to support analysis of the food–energy–water nexus. *Nature Sustainability*, 2(12), 1132–1141. <https://doi.org/10.1038/s41893-019-0418-8>
- Van Vuuren, D. P., Kriegler, E., O'Neill, B. C., Ebi, K. L., Riahi, K., Carter, T. R., et al. (2014). A new scenario framework for Climate Change Research: Scenario matrix architecture. *Climatic Change*, 122(3), 373–386. <https://doi.org/10.1007/s10584-013-0906-1>
- Vörösmarty, C. J., Green, P., Salisbury, J., & Lammers, R. B. (2000). Global water resources: Vulnerability from climate change and population growth. *Science*, 289(5477), 284–288. <https://doi.org/10.1126/science.289.5477.284>
- Wada, Y., & Bierkens, M. F. P. (2014). Sustainability of global water use: Past reconstruction and future projections. *Environmental Research Letters*, 9(10), 104003. <https://doi.org/10.1088/1748-9326/9/10/104003>
- Wessel, J., Iyer, G., Wild, T., Ou, Y., McJeon, H., & Lamontagne, J. (2024). Large ensemble exploration of global energy transitions under national emissions pledges. *Earth's Future*, 12(10), e2024EF004754. <https://doi.org/10.1029/2024EF004754>
- Weyant, J. (2014). Integrated assessment of climate change: State of the literature. *Journal of Benefit-Cost Analysis*, 5(3), 377–409. <https://doi.org/10.1515/jbca-2014-9002>
- Wise, M., Hodson, E. L., Mignone, B. K., Clarke, L., Waldhoff, S., & Luckow, P. (2015). An approach to computing marginal land use change carbon intensities for bioenergy in policy applications. *Energy Economics*, 50, 337–347. <https://doi.org/10.1016/j.eneco.2015.05.009>
- Woodard, D. L., Snyder, A., Lamontagne, J. R., Tebaldi, C., Morris, J., Calvin, K. V., et al. (2023). Scenario discovery analysis of drivers of solar and wind energy transitions through 2050. *Earth's Future*, 11(8), e2022EF003442. <https://doi.org/10.1029/2022EF003442>
- Zhang, Y., Ou, Y., Waldhoff, S., O'Neill, B., Iyer, G., Sampedro, J., & Casper, K. (2025). Long-term decarbonization impacts on residential energy security across income groups and US states. *Environmental Research Letters*, 20(5), 054040. <https://doi.org/10.1088/1748-9326/adca4c>

- Zhao, M., Khan, Z., Dorheim, K., & Vernon, C. (2024). Helios: An R package to process heating and cooling degrees for GCAM. *Journal of Open Source Software*, 9(94), 6033. <https://doi.org/10.21105/joss.06033>
- Zhao, M., Wild, T., & Vernon, C. (2023). Xanthos output dataset under ISIMIP3b selected CMIP6 scenarios [Dataset]. *MultiSector Dynamics - Living, Intuitive, Value-adding*, 1850–2100. <https://doi.org/10.57931/2280839>

References From the Supporting Information

- Hejazi, M., Edmonds, J., Chaturvedi, V., Davies, E., & Eom, J. (2013). Scenarios of global municipal water-use demand projections over the 21st century. *Hydrological Sciences Journal*, 58(3), 519–538. <https://doi.org/10.1080/02626667.2013.772301>
- Hejazi, M., Edmonds, J., Clarke, L., Kyle, P., Davies, E., Chaturvedi, V., et al. (2014). Long-term global water projections using six socio-economic scenarios in an integrated assessment modeling framework. *Technological Forecasting and Social Change*, 81, 205–226. <https://doi.org/10.1016/j.techfore.2013.05.006>
- Kyle, P., Hejazi, M., Kim, S., Patel, P., Graham, N., & Liu, Y. (2021). Assessing the future of global energy-for-water. *Environmental Research Letters*, 16(2), 024031. <https://doi.org/10.1088/1748-9326/abd8a9>